Heat and Work
27 Yes / No Questions

热与功：27 道判断题
中英文对照讲解

Each question includes the original English wording, a Chinese translation, an answer, a derivation, conditions of validity, and a worked check or example.

每道题都包含英文原题、中文翻译、答案、公式推导、适用条件，以及一个例子或检验。

Source quiz: *2 Quiz heat-work yes/no*, in *Thermodynamics, Ch. 1,2,3*. The 27 questions and displayed feedback were read in their original order.

题目来源为当前页面的“2 Quiz heat-work yes/no”。已按原顺序读取 27 道题及展开后的答案反馈。

The guide gives the quiz's intended answers while explaining conditions that the quiz leaves implicit. Questions 6, 11, 21, 22, and 25 deserve particular attention.

本讲义保留原题预期答案，同时说明原题未明确写出的适用条件。第 6、11、21、22、25 题尤其值得注意。

Prepared for Jiasheng Fu 29 October 2026

公式按教材形式分行排版；可通过 PDF 书签跳转至各题。

Answer key / 答案总表

Q	Answer	Q	Answer	Q	Answer	Q	Answer
1	Yes	8	No	15	Yes	22	Yes*
2	No	9	Yes	16	No	23	No
3	Yes*	10	No	17	Yes	24	Yes*
4	No	11	No*	18	Yes	25	Yes*
5	No	12	Yes*	19	No	26	No
6	No*	13	No	20	No	27	No
7	Yes	14	Yes	21	No*		

These are the quiz's intended answers. An asterisk highlights a qualification in the explanation. In particular, Question 6 is No as a general statement; equality can hold in special cases.

表中为原题预期答案。星号提示讲解中有重要适用条件。特别是第 6 题的一般答案为 No，但在特殊情况下两者可能相等。

Conventions / 符号约定

Heat into the system and work done on the system are positive. Positive work output is $W_{\text{out}} = -w$. The main balances below assume a closed system and negligible changes in bulk kinetic and potential energy unless otherwise stated.

系统吸热和外界对系统做功记为正。若把系统的对外输出功记为正，则 $W_{\text{out}} = -w$。除非另行说明，主要能量平衡针对封闭系统，且忽略整体动能及势能变化。

$$\boxed{\Delta U = q + w}, \qquad \delta w_{pV} = -p_{\text{ex}}\,\mathrm{d}V.$$

The symbols C_p, C_V denote total heat capacities unless a molar subscript is shown. ln is the natural logarithm. Temperatures used in gas equations are measured in kelvin.

除非注明摩尔下标，C_p, C_V 表示总热容。ln 是自然对数。气体公式中的温度使用开尔文。

The original wording and the quiz's displayed feedback were read; its seven listed source PDFs were not separately read. Derivations, translations, checks, and clarifications were prepared for this guide.

本讲义读取了题目原文与展开后的反馈；未单独读取其列出的七份来源 PDF。推导、翻译、例子和条件说明为本讲义整理。

QUESTION 1 / 27

Free expansion and boundary work

自由膨胀与边界功

Is the expansion work done by a gas expanding into a vacuum equal to zero?

气体向真空膨胀时，所做的膨胀功是否为零？

A. Yes / 是　B. No / 否

Answer and explanation / 答案与讲解

Yes

Boundary work requires the system boundary to move against an external load. In free expansion into a vacuum, the opposing external pressure is zero throughout, even though the gas volume increases.

边界功需要系统边界对抗外部负载发生位移。向真空自由膨胀时，即使气体体积增大，整个过程中阻碍膨胀的外压仍为零。

$$w_{pV} = -\int_{V_i}^{V_f} p_{\mathrm{ex}}\,\mathrm{d}V,$$
$$p_{\mathrm{ex}} = 0 \quad \Longrightarrow \quad \boxed{w_{pV} = 0}.$$

Conditions and common mistakes / 适用条件与易错点

This conclusion concerns expansion work. Other work interactions, if present, must be counted separately. Free expansion does not automatically mean the vessel is insulated: $q = 0$ requires an additional adiabatic condition.

这个结论针对膨胀功；若存在其他做功方式，需要另计。自由膨胀不自动意味着容器隔热：要得到 $q = 0$，还必须另加绝热条件。

Quick check / 例子与检验

For an insulated ideal-gas free expansion with no other work, $q = w = 0$ and therefore $\Delta U = 0$. The entropy can nevertheless increase.

例如，绝热理想气体自由膨胀且无其他功时，$q = w = 0$，因此 $\Delta U = 0$；但熵仍可以增加。

QUESTION 2 / 27

Ideal-gas energy at fixed temperature

恒温时理想气体的内能

Is the internal energy of a perfect gas dependent on the volume it occupies at constant temperature?

在温度恒定时，理想气体的内能是否取决于它所占的体积？

A. No / 否　B. Yes / 是

Answer and explanation / 答案与讲解

No

For a fixed amount and composition of a perfect gas, U is a function of temperature alone. Increasing its volume at the same temperature does not change its equilibrium internal energy.

对于物质的量和组成固定的理想气体，内能 U 只依赖温度。在相同温度下增大体积，不会改变其平衡态内能。

$$\begin{aligned} \mathrm{d}U &= C_V(T)\,\mathrm{d}T, \\ \left(\frac{\partial U}{\partial V}\right)_{T,n} &= 0, \\ \Delta U &= 0 \qquad (T_f = T_i). \end{aligned}$$

Conditions and common mistakes / 适用条件与易错点

The ideal-gas model neglects intermolecular forces. Real gases can have volume-dependent potential energy. Molecules may still possess rotational and vibrational internal energy; those contributions can depend on temperature.

理想气体模型忽略分子间作用；真实气体的势能可能依赖体积。分子仍可具有转动和振动内能，这些贡献可以随温度改变。

Quick check / 例子与检验

If an ideal gas doubles its volume while remaining at 300 K, its entropy changes, but its internal energy does not.

例如，理想气体在 300 K 下体积变为两倍时，熵会改变，但内能不变。

QUESTION 3 / 27

Reversible expansion and maximum work

可逆膨胀与最大输出功

Does a system perform the maximum possible expansion work when the process is carried out reversibly?

系统进行可逆过程时，是否能做出最大可能的膨胀功？

A. No / 否　B. Yes / 是

Answer and explanation / 答案与讲解

Yes, under specified constraints

For given equilibrium endpoints and the same thermal constraints, reversible expansion yields the maximum possible work output. In the reversible limit the external pressure is infinitesimally below the gas pressure throughout an expansion.

在给定平衡始末态并保持相同热条件时，可逆膨胀给出最大对外输出功。在可逆极限中，膨胀过程的外压每一步都比气体压强小无穷小。

$$
\begin{aligned}
W_{\mathrm{out}} &= \int_{V_i}^{V_f} p_{\mathrm{ex}}\,\mathrm{d}V, \\
W_{\mathrm{out}} &\le W_{\mathrm{out,rev}}, \\
W_{\mathrm{out,rev}} &= nRT\ln\!\left(\frac{V_f}{V_i}\right) \quad \text{(isothermal ideal gas)}.
\end{aligned}
$$

Conditions and common mistakes / 适用条件与易错点

The comparison is meaningful only after fixing endpoints and constraints, such as a common thermal reservoir. The signed work on the system is $w = -W_{\mathrm{out}}$, so its reversible value is most negative, not numerically largest.

比较之前必须固定始末态与约束，例如使用同一恒温热库。若把外界对系统做功记为 $w = -W_{\mathrm{out}}$，可逆值是最负的，而不是数值最大的。

Quick check / 例子与检验

For one mole at $300\,\mathrm{K}$ expanding from V to $2V$, $W_{\mathrm{out,rev}} = RT\ln 2 \approx 1.73\,\mathrm{kJ}$.

例如，一摩尔理想气体在 $300\,\mathrm{K}$ 从 V 膨胀到 $2V$，可逆输出功为 $RT\ln 2 \approx 1.73\,\mathrm{kJ}$。

QUESTION 4 / 27

Internal energy is a state function

内能是状态函数

Is internal energy (U) classified as a path function?

内能 U 是否属于路径函数？

A. Yes / 是　B. No / 否

Answer and explanation / 答案与讲解

No

The internal energy of a specified equilibrium state has a definite value. Its change between two states is independent of how the system travels between them.

一个指定平衡态的内能具有确定值。两个状态之间的内能变化，只取决于这两个状态，与系统经过的路径无关。

$$\Delta U = U_f - U_i,$$
$$\oint \mathrm{d}U = 0,$$
$$\Delta U = q_1 + w_1 = q_2 + w_2.$$

Conditions and common mistakes / 适用条件与易错点

Heat and work are transfers and may differ along paths 1 and 2. The sum remains the same for the same initial and final states. The differential $\mathrm{d}U$ is exact; heat and work are conventionally written δq and δw.

热和功是能量传递，在路径 1 与路径 2 上可以不同；只要始末态相同，它们的和就相同。$\mathrm{d}U$ 是全微分；热与功的微小传递通常写成 δq 和 δw。

Quick check / 例子与检验

An ideal gas returning to its starting state has $\Delta U = 0$, regardless of whether parts of the cycle are reversible.

例如，理想气体回到初始状态后，$\Delta U = 0$，与循环中的部分过程是否可逆无关。

QUESTION 5 / 27

Work depends on the path

功取决于过程路径

Is work (w) classified as a state function?

功 w 是否属于状态函数？

A. Yes / 是　B. No / 否

Answer and explanation / 答案与讲解

No

Work is energy transferred through a generalized force and displacement. Different external pressure histories can connect identical equilibrium states while transferring different amounts of work.

功是通过广义力与位移传递的能量。不同的外压变化过程可以连接相同的平衡始末态，但传递的功不同。

$$w = -\int_{V_i}^{V_f} p_{\mathrm{ex}}\,\mathrm{d}V,$$

$$w_{\mathrm{rev},T} = -nRT\ln\left(\frac{V_f}{V_i}\right),$$

$$w_{\mathrm{free}} = 0.$$

Conditions and common mistakes / 适用条件与易错点

These last two results can apply to the same initial and final ideal-gas states, but require different physical arrangements. Work is not stored in the system as a state property.

最后两式可以对应理想气体相同的始末态，但需要不同的实验安排。功并不是系统内部储存的状态性质。

Quick check / 例子与检验

In an isothermal ideal-gas expansion with $V_f > V_i$, reversible work on the system is negative, whereas vacuum expansion gives zero boundary work.

例如，理想气体等温膨胀且 $V_f > V_i$ 时，可逆过程的对系统功为负，向真空膨胀的边界功却为零。

QUESTION 6 / 27

Constant-volume heat and enthalpy

定容热与焓变

Is the energy transferred as heat at constant volume (q_V) equal to the change in enthalpy (ΔH) when no non-expansion work is done?

在没有非膨胀功时，定容传入的热量 q_V 是否等于焓变 ΔH？

A. No / 否　B. Yes / 是

Answer and explanation / 答案与讲解

No, not in general

For a closed system at fixed volume with no non-expansion work, the first law gives $q_V = \Delta U$. Enthalpy includes the additional state term pV, so q_V is not generally equal to ΔH.

封闭系统定容且不做非膨胀功时，第一定律给出 $q_V = \Delta U$。焓还包含状态项 pV，所以 q_V 一般不等于 ΔH。

$$\begin{aligned} \Delta V = 0 \quad &\Longrightarrow \quad w_{pV} = 0, \\ q_V &= \Delta U, \\ \Delta H &= \Delta U + \Delta(pV) = q_V + V\Delta p. \end{aligned}$$

Conditions and common mistakes / 适用条件与易错点

The quiz's No means "not always": equality is possible in a special case with $\Delta(pV) = 0$. The endpoint pressures must be defined; a system with non-expansion work requires an extra work term.

原题的 No 应理解为"一般不成立"：特殊情况下若 $\Delta(pV) = 0$，两者仍可能相等。这里要求始末态的压强有定义；若存在非膨胀功，还需加入额外功项。

Quick check / 例子与检验

For fixed-n ideal gas heated at constant V, $\Delta H - \Delta U = nR\Delta T$, which is nonzero for a nonzero temperature change.

例如，固定物质的量的理想气体定容升温时，$\Delta H - \Delta U = nR\Delta T$；只要温度变化非零，两者便不同。

QUESTION 7 / 27

Constant-pressure heat and enthalpy

恒压热与焓变

At constant pressure, is the change in enthalpy (ΔH) equal to the heat transferred (q_p) provided no non-expansion work is done?

恒压且没有非膨胀功时，焓变 ΔH 是否等于传递的热量 q_p？

A. Yes / 是　B. No / 否

Answer and explanation / 答案与讲解

Yes, with the stated work restriction

For a closed system at constant pressure, with pressure-volume work only and mechanical balance against that pressure, the heat absorbed equals the enthalpy change.

封闭系统恒压，且只做压力—体积功，并与该外压机械平衡时，系统吸收的热量等于焓变。

$$\begin{aligned} H &= U + pV, \\ \Delta U &= q_p - p\Delta V, \\ \Delta H &= \Delta U + p\Delta V = \boxed{q_p}. \end{aligned}$$

Conditions and common mistakes / 适用条件与易错点

Constant pressure alone does not rule out electrical or shaft work. If non-expansion work w_{other} is done on the system, then $\Delta H = q_p + w_{\text{other}}$ under the same pressure assumptions.

恒压本身不能排除电功或轴功。若外界对系统做非膨胀功 w_{other}，在相同压力假设下应写成 $\Delta H = q_p + w_{\text{other}}$。

Quick check / 例子与检验

If the system releases $50\,\text{kJ}$ as heat with no other work, $q_p = -50\,\text{kJ}$ and $\Delta H = -50\,\text{kJ}$.

例如，系统没有其他功而放热 $50\,\text{kJ}$ 时，$q_p = -50\,\text{kJ}$，故 $\Delta H = -50\,\text{kJ}$。

QUESTION 8 / 27

Comparing heat capacities

比较定压和定容热容

Is the constant-pressure heat capacity (C_p) of a gas smaller than its constant-volume heat capacity (C_V)?

气体的定压热容 C_p 是否小于定容热容 C_V？

A. No / 否　B. Yes / 是

Answer and explanation / 答案与讲解

No

For an ideal gas, raising temperature at constant pressure also requires expansion work. The difference between its total heat capacities is exactly nR for fixed amount and composition.

理想气体恒压升温时还要对外膨胀做功。固定物质的量和组成时，总定压热容与总定容热容之差恰好为 nR。

$$C_p = \left(\frac{\partial H}{\partial T}\right)_p, \qquad C_V = \left(\frac{\partial U}{\partial T}\right)_V,$$
$$H = U + nRT,$$
$$\boxed{C_p - C_V} = nR > 0 \quad \text{(ideal gas)}.$$

Conditions and common mistakes / 适用条件与易错点

For a stable simple compressible material, the more general relation is $C_p - C_V = TV\alpha^2/\kappa_T \geq 0$ when the isothermal compressibility κ_T is positive. Equality can occur; nR is not the general real-gas result.

对于稳定的简单可压缩物质，更一般的关系为 $C_p - C_V = TV\alpha^2/\kappa_T \geq 0$，其中等温压缩率 $\kappa_T > 0$。两者有时可以相等；nR 并非真实气体的一般差值。

Quick check / 例子与检验

For a classical monatomic ideal gas, $C_{V,\mathrm{m}} = 3R/2$ and $C_{p,\mathrm{m}} = 5R/2$.

例如，经典单原子理想气体的 $C_{V,\mathrm{m}} = 3R/2$，而 $C_{p,\mathrm{m}} = 5R/2$。

QUESTION 9 / 27

Exothermic means negative enthalpy change

放热对应负焓变

Does an exothermic chemical process taking place at constant pressure correspond to a negative change in enthalpy ($\Delta H < 0$)?

恒压下发生的放热化学过程是否对应负的焓变 $\Delta H < 0$？

A. No / 否　B. Yes / 是

Answer and explanation / 答案与讲解

Yes, with pressure-volume work only

An exothermic process transfers heat out of the system. Under the chemistry sign convention, that heat is negative; at constant pressure with no additional work, it equals the enthalpy change.

放热过程把热量从系统传向环境。按照化学符号约定，这个热量为负；在恒压且没有额外功时，它等于焓变。

$$\begin{aligned} q_{p,\mathrm{sys}} &< 0, \\ \Delta H_{\mathrm{sys}} &= q_{p,\mathrm{sys}} < 0, \\ q_{\mathrm{surr}} &= -q_{p,\mathrm{sys}} > 0. \end{aligned}$$

Conditions and common mistakes / 适用条件与易错点

The sign is for the forward process as written. Reversing the reaction reverses ΔH. Additional non-expansion work can break the simple equality between q_p and ΔH.

此符号对应按题目写出的正向过程。反应方向反转时，ΔH 的符号也反转。若存在额外的非膨胀功，$q_p = \Delta H$ 的简单关系可能不成立。

Quick check / 例子与检验

For a reaction releasing 125 kJ per mole of reaction, $\Delta_r H = -125\,\mathrm{kJ\,mol^{-1}}$ under the stated conditions.

例如，在上述条件下，反应每摩尔反应进度放出 125 kJ，则 $\Delta_r H = -125\,\mathrm{kJ\,mol^{-1}}$。

QUESTION 10 / 27

Total heat capacity is extensive

总热容是广度性质

Is the total heat capacity (C_V) of a sample an intensive property?

样品的总定容热容 C_V 是否属于强度性质？

A. Yes / 是　B. No / 否

Answer and explanation / 答案与讲解

No

An extensive property scales with system size when independent, otherwise identical portions are combined. Total heat capacity therefore doubles when the amount of the same homogeneous material doubles.

广度性质会在其他状态相同的独立部分合并时随系统大小缩放。因此，均匀材料的量增加一倍，总热容也增加一倍。

$$\begin{aligned} C_V &= nC_{V,\mathrm{m}}, \\ n \longrightarrow 2n \quad &\Longrightarrow \quad C_V \longrightarrow 2C_V, \\ [C_V] &= \mathrm{J\,K^{-1}}. \end{aligned}$$

Conditions and common mistakes / 适用条件与易错点

The comparison holds at the same temperature, density, composition, and phase, with negligible finite-size effects. Molar heat capacity $C_{V,\mathrm{m}}$ is intensive, unlike total C_V.

比较要求温度、密度、组成和物相相同，并忽略有限尺寸效应。摩尔定容热容 $C_{V,\mathrm{m}}$ 是强度性质，与总热容 C_V 不同。

Quick check / 例子与检验

If one mole has $C_V = 20\,\mathrm{J\,K^{-1}}$, two moles in the same state have $C_V = 40\,\mathrm{J\,K^{-1}}$.

例如，同一状态下一摩尔样品的 $C_V = 20\,\mathrm{J\,K^{-1}}$，则两摩尔样品的总热容为 $40\,\mathrm{J\,K^{-1}}$。

QUESTION 11 / 27

Reversibility as an ideal limit

可逆过程是理想极限

Can a truly thermodynamic reversible process be fully completed within a finite time in a laboratory?

真正的热力学可逆过程能否在实验室中有限时间内完全完成？

A. Yes / 是　B. No / 否

Answer and explanation / 答案与讲解

No, in the usual laboratory limit

The quiz's intended answer is No: an ordinary finite-rate laboratory process has finite driving forces and friction or other dissipation. A reversible process is approached by making those losses vanish.

原题预期答案是 No：普通实验室中的有限速率过程具有有限驱动力，以及摩擦或其他耗散。要趋近可逆过程，就需要使这些损失趋于零。

$$\begin{aligned} \Delta S_{\mathrm{tot}} &= S_{\mathrm{gen}} \geq 0, \\ S_{\mathrm{gen}} &= 0 \quad \text{for a reversible process,} \\ S_{\mathrm{gen}} &> 0 \quad \text{for an irreversible process.} \end{aligned}$$

Conditions and common mistakes / 适用条件与易错点

Quasistatic alone is not sufficient: a slowly moving piston with friction still produces entropy. "Infinite time" describes the usual idealized limiting construction, not a universal theorem for every conceivable ideal frictionless model.

仅仅缓慢或准静态还不够：有摩擦的活塞即使运动很慢仍会产生熵。"需要无限时间"描述通常的理想极限构造，并非对任何可想象的无摩擦理想模型都成立的普遍定理。

Quick check / 例子与检验

As the temperature difference driving heat transfer is reduced, entropy generation can be reduced; finite thermal conductance then typically makes a fixed heat transfer slower.

例如，减小传热温差可以减少熵产生；若导热能力有限，要传递固定热量通常就需要更长时间。

QUESTION 12 / 27

First law for an isolated system

孤立系统形式的第一定律

Is the First Law of thermodynamics equivalent to stating that the internal energy of an isolated system is constant?

热力学第一定律是否等价于“孤立系统的内能保持不变”这一表述？

A. No / 否　B. Yes / 是

Answer and explanation / 答案与讲解

Yes, as the isolated-system form

Energy conservation implies that the total energy of an isolated system is constant. In the standard thermodynamic setting, the system has no matter, heat, or work transfer across its boundary, and changes in bulk kinetic and potential energy are neglected.

能量守恒意味着孤立系统的总能量保持不变。在通常热力学设置中，孤立系统边界没有物质、热或功的传递，并忽略整体动能和势能变化。

$$\begin{aligned} \Delta E_{\text{total}} &= 0, \\ \Delta U &= q + w, \\ q = w = 0 \quad &\Longrightarrow \quad \boxed{\Delta U = 0}. \end{aligned}$$

Conditions and common mistakes / 适用条件与易错点

For a general moving or externally situated system, the conserved quantity is total energy, $E = U + E_{\text{kin}} + E_{\text{pot}} + \cdots$. The isolated-system statement is a standard form of the first law; open-system balances require matter-carried energy terms.

对于整体运动或处于外势场中的一般系统，守恒量是总能量 $E = U + E_{\text{kin}} + E_{\text{pot}} + \cdots$。孤立系统表述是第一定律的标准形式；开放系统的能量平衡还需要考虑物质携带的能量。

Quick check / 例子与检验

An insulated rigid vessel can contain an internal chemical reaction with $\Delta U = 0$ even though its temperature and composition change.

例如，绝热刚性容器内可以发生化学反应，温度和组成虽发生改变，整个孤立系统仍满足 $\Delta U = 0$。

QUESTION 13 / 27

Endothermic means positive enthalpy change

吸热对应正焓变

Does an endothermic reaction occurring at constant pressure result in a decrease in enthalpy ($\Delta H < 0$)?

恒压下的吸热反应是否会导致焓减小，即 $\Delta H < 0$？

A. No / 否　B. Yes / 是

Answer and explanation / 答案与讲解

No

An endothermic reaction absorbs heat from the surroundings. In the usual closed-system, constant-pressure setting with no non-expansion work, that means a positive enthalpy change.

吸热反应从环境吸收热量。在通常的封闭系统恒压条件下，若没有非膨胀功，这意味着正的焓变。

$$\begin{aligned} q_{p,\mathrm{sys}} &> 0, \\ \Delta H_{\mathrm{sys}} = q_{p,\mathrm{sys}} &> 0, \\ q_{\mathrm{surr}} = -q_{p,\mathrm{sys}} &< 0. \end{aligned}$$

Conditions and common mistakes / 适用条件与易错点

A negative ΔH describes the reverse, exothermic direction under the same convention. As in Question 7, additional work requires a modified balance.

按同样的约定，负的 ΔH 对应反向的放热过程。与第 7 题一样，若有额外功，必须修正能量平衡。

Quick check / 例子与检验

If melting absorbs $6.0\,\mathrm{kJ}$ at constant pressure and no other work occurs, $\Delta H = +6.0\,\mathrm{kJ}$ for that sample.

例如，在恒压且没有其他功时，样品熔化吸收 $6.0\,\mathrm{kJ}$，该样品的 $\Delta H = +6.0\,\mathrm{kJ}$。

QUESTION 14 / 27

The sign of heat entering the system

传入系统的热量符号

In standard thermodynamic conventions, is energy transferred to a system as heat assigned a positive sign ($q > 0$)?

按通常的热力学符号约定，以热的形式传入系统的能量是否记为正值 $q > 0$？

A. Yes / 是　B. No / 否

Answer and explanation / 答案与讲解

Yes

In the chemistry convention used throughout this guide, heat entering the system is positive and heat leaving is negative. Work done on the system is also positive.

本讲义使用化学符号约定：热进入系统为正，热离开系统为负；外界对系统做功也记为正。

$$\begin{aligned} \Delta U &= q + w, \\ q > 0 &: \text{ heat enters the system,} \\ q < 0 &: \text{ heat leaves the system.} \end{aligned}$$

Conditions and common mistakes / 适用条件与易错点

Positive heat alone does not guarantee $\Delta U > 0$: simultaneous work output may exceed the heat input. The sign applies to the system; the same heat has the opposite sign for the surroundings.

只有热量为正，并不能保证 $\Delta U > 0$：系统同时对外做的功可能大于吸收的热。符号从系统角度确定；同一笔热量对环境具有相反符号。

Quick check / 例子与检验

If $q = +100\,\mathrm{J}$ and $w = -150\,\mathrm{J}$, then $\Delta U = -50\,\mathrm{J}$.

例如，若 $q = +100\,\mathrm{J}$ 而 $w = -150\,\mathrm{J}$，则 $\Delta U = -50\,\mathrm{J}$。

QUESTION 15 / 27

Internal energy over a complete cycle

完整循环的内能变化

In any thermodynamic cyclic process, is the total net change in internal energy (ΔU) equal to zero?

在任何热力学循环中，内能的净变化 ΔU 是否等于零？

A. No / 否　B. Yes / 是

Answer and explanation / 答案与讲解

Yes

A thermodynamic cycle returns the system to its initial state. Since internal energy is a state function, its net change over the complete cycle must vanish.

热力学循环使系统回到初始状态。由于内能是状态函数，完整循环的内能净变化必定为零。

$$U_f = U_i,$$
$$\Delta U_{\text{cycle}} = \oint \mathrm{d}U = 0,$$
$$q_{\text{net}} + w_{\text{net}} = 0.$$

Conditions and common mistakes / 适用条件与易错点

Zero net internal-energy change does not mean that heat and work are each zero. A heat engine can absorb and reject heat and deliver net work while returning its working substance to the original state.

内能净变化为零，不代表热与功各自为零。热机可以吸热、放热并输出净功，同时使工质回到原来的状态。

Quick check / 例子与检验

If the net heat absorbed over a cycle is $200\,\mathrm{J}$, the net work on the system is $-200\,\mathrm{J}$ in this convention.

例如，循环净吸热 $200\,\mathrm{J}$ 时，按本讲义约定，外界对系统的净功为 $-200\,\mathrm{J}$。

QUESTION 16 / 27

Standard pressure: bar versus atmosphere

标准压力：巴与标准大气压

Is the standard pressure for reporting thermodynamic data currently defined as 1 *atm* (101.325 *kPa)?*

目前报告热力学数据时，标准压力是否定义为 1 atm（101.325 kPa）？

A. Yes / 是　B. No / 否

Answer and explanation / 答案与讲解

No

The modern IUPAC standard pressure is 1 bar = 100 kPa. The standard atmosphere is a distinct pressure unit equal to 101.325 kPa. Older thermodynamic tables can use 1 atm. [R1]

现代 IUPAC 标准压力为 1 bar = 100 kPa。标准大气压是另一单位，等于 101.325 kPa。较早的热力学数据表可能采用 1 atm。[R1]

$$\begin{aligned} p^\circ &= 1\,\mathrm{bar} = 100\,000\,\mathrm{Pa}, \\ 1\,\mathrm{atm} &= 101\,325\,\mathrm{Pa} \\ &= 1.01325\,\mathrm{bar}. \end{aligned}$$

Conditions and common mistakes / 适用条件与易错点

The two pressures are close but not identical. “Standard” can mean different reference conditions in other disciplines, so check the definition attached to a table or problem. Standard-state pressure alone does not select a temperature.

两个压力很接近，但并不相同。其他学科的“标准条件”可能采用不同参考值，因此应查看题目或数据表的定义。标准态压力本身也不指定温度。

Quick check / 例子与检验

A report at 298.15 K and 1 bar uses a pressure lower than 1 atm by 1.325 kPa.

例如，298.15 K、1 bar 的报告条件，其压强比 1 atm 低 1.325 kPa。

QUESTION 17 / 27

Defining constant-volume heat capacity

定容热容的定义

Is constant-volume heat capacity mathematically defined as $C_V = \left(\frac{\partial U}{\partial T}\right)_V$*?*

定容热容的数学定义是否为 $C_V = (\partial U/\partial T)_V$?

A. Yes / 是 B. No / 否

Answer and explanation / 答案与讲解

Yes

At fixed volume and composition, C_V is the rate of change of internal energy with temperature. It is a derivative, so it refers to the local slope and can itself vary with temperature.

固定体积和组成时，C_V 是内能对温度的变化率。它是偏导数，表示局部斜率，因此热容自身也可以随温度改变。

$$C_V \equiv \left(\frac{\partial U}{\partial T}\right)_{V,n},$$
$$\Delta U = \int_{T_i}^{T_f} C_V(T)\,\mathrm{d}T \quad (V, n \text{ fixed}),$$
$$C_{V,\mathrm{m}} = \frac{C_V}{n}.$$

Conditions and common mistakes / 适用条件与易错点

The familiar statement $q_V = \Delta U$ additionally requires a closed system with no non-expansion work. The heat-capacity definition itself is a state-property derivative; it does not assert that every fixed-volume experiment has no electrical work.

常见的 $q_V = \Delta U$ 还要求系统封闭且没有非膨胀功。热容定义本身是状态性质的偏导数，并不表示所有定容实验都没有电功。

Quick check / 例子与检验

If total C_V is constant at $20\,\mathrm{J\,K^{-1}}$ and temperature rises by $5\,\mathrm{K}$, then $\Delta U = 100\,\mathrm{J}$.

例如，总 $C_V = 20\,\mathrm{J\,K^{-1}}$ 且近似不变时，升温 $5\,\mathrm{K}$ 给出 $\Delta U = 100\,\mathrm{J}$。

QUESTION 18 / 27

Isothermal ideal-gas expansion

理想气体等温膨胀

For a perfect gas undergoing an isothermal expansion, is the change in internal energy (ΔU) equal to zero?

理想气体进行等温膨胀时，其内能变化 ΔU 是否为零？

A. No / 否　B. Yes / 是

Answer and explanation / 答案与讲解

Yes

For a fixed amount of ideal gas, internal energy depends only on temperature. An isothermal expansion therefore has identical initial and final internal energies.

固定物质的量的理想气体，其内能只依赖温度。因此，等温膨胀的始末态内能相同。

$$\Delta U = \int_{T_i}^{T_f} C_V(T)\,\mathrm{d}T,$$
$$T_f = T_i \quad \Longrightarrow \quad \Delta U = 0,$$
$$0 = q + w \quad \Longrightarrow \quad q = -w.$$

Conditions and common mistakes / 适用条件与易错点

The result applies to reversible and irreversible paths between the same ideal-gas endpoints. It does not imply that q and w vanish individually. For a real gas, isothermal expansion can change internal energy.

此结论适用于连接相同理想气体始末态的可逆或不可逆路径，但不意味着 q 和 w 各自为零。真实气体的等温膨胀则可能改变内能。

Quick check / 例子与检验

In a reversible expansion from V_i to $2V_i$, $w = -nRT\ln 2$ and $q = +nRT\ln 2$.

例如，从 V_i 可逆膨胀到 $2V_i$ 时，$w = -nRT\ln 2$，而 $q = +nRT\ln 2$。

QUESTION 19 / 27

Work in a free expansion

自由膨胀是否对外做功

Does free expansion against zero external pressure ($p_{ex} = 0$) do work on the surroundings?

在外压为零（$p_{\mathrm{ex}} = 0$）的自由膨胀中，气体是否对环境做功？

A. No / 否　B. Yes / 是

Answer and explanation / 答案与讲解

No, for expansion work

With zero external opposing pressure, the moving gas boundary does no pressure-volume work on the surroundings. This restates Question 1 from the surroundings' perspective.

外部阻碍压强为零时，气体边界不对环境做压力—体积功。这是第 1 题从环境角度提出的同一问题。

$$W_{\mathrm{out},pV} = \int_{V_i}^{V_f} p_{\mathrm{ex}}\,\mathrm{d}V,$$
$$p_{\mathrm{ex}} = 0 \quad \Longrightarrow \quad W_{\mathrm{out},pV} = 0,$$
$$w_{pV} = -W_{\mathrm{out},pV} = 0.$$

Conditions and common mistakes / 适用条件与易错点

A gas's internal pressure is not the opposing pressure in the boundary-work integral. The answer concerns the expansion-work mechanism; separately imposed electrical or other work is not excluded merely by $p_{\mathrm{ex}} = 0$.

边界功积分中使用的阻碍压强不是气体的内部压强。答案针对膨胀功；仅凭 $p_{\mathrm{ex}} = 0$ 并不能排除另外施加的电功或其他功。

Quick check / 例子与检验

A gas can move into an evacuated compartment and reach a larger volume without raising a weight or moving a loaded piston.

例如，气体可以进入抽空的隔室而增加体积，但不需要抬升重物或推动受负载的活塞。

QUESTION 20 / 27

Enthalpy versus energy in gas reactions

气体反应中的焓变与内能变

For a chemical reaction involving gases at constant temperature, is ΔH always identical to ΔU regardless of changes in the amount of gas molecules (Δn_g)?

对于恒温下涉及气体的化学反应，不论气体物质的量的变化 Δn_{g} 如何，ΔH 是否总与 ΔU 相同？

A. No / 否　B. Yes / 是

Answer and explanation / 答案与讲解

No

The difference is the change in the state function pV. For ideal gases at a common temperature, a change in gas amount contributes a term proportional to RT.

两者之差等于状态项 pV 的变化。若气体为理想气体且处于同一温度，气体物质的量变化会产生一个与 RT 成正比的项。

$$\Delta H = \Delta U + \Delta(pV),$$
$$\Delta H \simeq \Delta U + \Delta n_{\mathrm{g}} RT,$$
$$\Delta n_{\mathrm{g}} = n_{\mathrm{gas,products}} - n_{\mathrm{gas,reactants}}.$$

Conditions and common mistakes / 适用条件与易错点

The second relation assumes ideal gases, a common initial and final temperature, and negligible changes in condensed-phase pV contributions. With reaction enthalpy per mole of reaction, use $\Delta\nu_{\mathrm{g}} RT$ instead. If $\Delta n_{\mathrm{g}} = 0$, equality holds within this approximation.

第二式要求气体近似理想、始末温度相同，并忽略凝聚相 pV 的变化。若使用每摩尔反应进度的反应焓，则应写 $\Delta\nu_{\mathrm{g}} RT$。在该近似下，若 $\Delta n_{\mathrm{g}} = 0$，两者相等。

Quick check / 例子与检验

For $\mathrm{A(g)} \rightarrow 2\mathrm{B(g)}$ per mole of reaction, $\Delta\nu_{\mathrm{g}} = 1$ and $\Delta_r H - \Delta_r U = RT$.

例如，对 $\mathrm{A(g)} \rightarrow 2\mathrm{B(g)}$，每摩尔反应进度有 $\Delta\nu_{\mathrm{g}} = 1$，所以 $\Delta_r H - \Delta_r U = RT$。

QUESTION 21 / 27

Standard state does not fix temperature

标准态不规定固定温度

Does the official definition of a substance's standard state mandate that its temperature must strictly be 298.15 *K?*

一种物质的标准态的正式定义，是否要求温度必须严格为 $298.15\,\mathrm{K}$？

A. No / 否　B. Yes / 是

Answer and explanation / 答案与讲解

No

The superscript ∘ specifies a standard reference state, while temperature must still be stated. Tables often use $298.15\,\mathrm{K}$, but standard quantities can be defined at other temperatures. [R2]

上标 ∘ 表示标准参考态，但温度仍需另行指定。数据表经常使用 $298.15\,\mathrm{K}$，标准热力学量也可以在其他温度定义。[R2]

$$\begin{aligned}\Delta_r H^\circ &= \Delta_r H^\circ(T),\\ \Delta_r G^\circ &= \Delta_r G^\circ(T),\\ p^\circ &= 1\,\mathrm{bar}\quad \text{(IUPAC standard pressure)}.\end{aligned}$$

Conditions and common mistakes / 适用条件与易错点

The quiz feedback oversimplifies by saying standard state means only 1 bar. A gas standard state is the ideal-gas reference at standard pressure; solute standard states can use specified standard molality or concentration. The definition depends on the phase and convention.

原反馈称标准态"只指 1 bar"过于简单。气体标准态是标准压力下的理想气体参考态；溶质标准态也可能以标准质量摩尔浓度或标准物质的量浓度定义。具体定义取决于物相与约定。

Quick check / 例子与检验

Both $\Delta_r H^\circ(298.15\,\mathrm{K})$ and $\Delta_r H^\circ(350\,\mathrm{K})$ are legitimate when the standard-state conventions are specified.

例如，只要标准态约定明确，$\Delta_r H^\circ(298.15\,\mathrm{K})$ 和 $\Delta_r H^\circ(350\,\mathrm{K})$ 都可以定义。

QUESTION 22 / 27

Heat capacity near absolute zero

接近绝对零度时的热容

Does the heat capacity of every pure substance approach zero as its thermodynamic temperature approaches absolute zero ($T = 0$)?

每一种纯物质的热容是否都在热力学温度趋近绝对零度时趋于零？

A. Yes / 是　B. No / 否

Answer and explanation / 答案与讲解

Yes in the usual equilibrium model

The quiz expects Yes. For an ordinary equilibrium phase in the standard third-law setting, thermal excitations disappear as $T \to 0$ and the heat capacity approaches zero. A low-temperature Debye solid provides a concrete example. [R3]

原题预期答案是 Yes。在通常第三定律适用的平衡态条件下，当 $T \to 0$ 时，普通物质的热激发逐渐消失，热容趋于零。低温德拜固体是一个具体例子。[R3]

$$C_{V,\mathrm{m}} \simeq aT^3 \quad \text{(Debye lattice contribution)},$$
$$\lim_{T\to 0} C_{V,\mathrm{m}} = 0,$$
$$S(T) - S(0) = \int_0^T \frac{C(T')}{T'}\,\mathrm{d}T'.$$

Conditions and common mistakes / 适用条件与易错点

The word “every” needs care: phase transitions, nonequilibrium states, and unusual limiting paths can complicate the limit. A vanishing heat capacity does not prove zero residual entropy; frozen configurational disorder can give $S(0) > 0$ while $C \to 0$.

“每一种”需要谨慎：相变、非平衡态和特殊极限路径都可能使问题更复杂。热容趋于零并不能证明剩余熵为零；被冻结的构型无序可以使 $S(0) > 0$，同时 $C \to 0$。

Quick check / 例子与检验

If $C_{\mathrm{m}} = aT^3$, then $S_{\mathrm{m}}(T) - S_{\mathrm{m}}(0) = aT^3/3 = C_{\mathrm{m}}(T)/3$.

例如，若 $C_{\mathrm{m}} = aT^3$，则 $S_{\mathrm{m}}(T) - S_{\mathrm{m}}(0) = aT^3/3 = C_{\mathrm{m}}(T)/3$。

QUESTION 23 / 27

Molar heat capacity is intensive

摩尔热容是强度性质

Is molar heat capacity (C_m) classified as an extensive property?

摩尔热容 C_{m} 是否属于广度性质？

A. Yes / 是　B. No / 否

Answer and explanation / 答案与讲解

No

Molar heat capacity is the total heat capacity divided by the amount of substance. Scaling the same homogeneous sample up or down changes both numerator and denominator by the same factor.

摩尔热容等于总热容除以物质的量。同一均匀样品放大或缩小时，分子和分母按相同倍数变化。

$$C_{\mathrm{m}} = \frac{C}{n},$$
$$\frac{\lambda C}{\lambda n} = C_{\mathrm{m}},$$
$$[C_{\mathrm{m}}] = \mathrm{J\,mol^{-1}\,K^{-1}}.$$

Conditions and common mistakes / 适用条件与易错点

The intensive statement compares the same phase and thermodynamic state. Molar heat capacity can vary with temperature, pressure, composition, and phase; intensive does not mean constant under all conditions.

强度性质的比较要求物相和热力学状态相同。摩尔热容仍可随温度、压强、组成和物相变化；“强度”并不意味着在所有条件下都是常数。

Quick check / 例子与检验

If $C = 50\,\mathrm{J\,K^{-1}}$ for $2\,\mathrm{mol}$, then $C_{\mathrm{m}} = 25\,\mathrm{J\,mol^{-1}\,K^{-1}}$.

例如，$2\,\mathrm{mol}$ 样品的总热容为 $50\,\mathrm{J\,K^{-1}}$，则摩尔热容为 $25\,\mathrm{J\,mol^{-1}\,K^{-1}}$。

QUESTION 24 / 27

The pressure-volume work differential

压力—体积功微分式

Does the differential expression for expansion work, $dw = -p_{ex}dV$, apply to both reversible and irreversible expansions?

膨胀功的微分表达式 $\mathrm{d}w = -p_{\mathrm{ex}}\,\mathrm{d}V$ 是否同时适用于可逆和不可逆膨胀？

A. Yes / 是　B. No / 否

Answer and explanation / 答案与讲解

Yes, for the stated boundary model

For a piston or boundary acted on by a well-defined uniform external pressure, the pressure-volume work on the system is $-p_{\mathrm{ex}}\mathrm{d}V$. This applies to reversible and irreversible motion.

对于受到均匀且定义明确的外压作用的活塞或边界，外界对系统做的体积功为 $-p_{\mathrm{ex}}\mathrm{d}V$。这一式子既适用于可逆运动，也适用于不可逆运动。

$$\begin{aligned} \delta w_{pV} &= -p_{\mathrm{ex}}\,\mathrm{d}V, \\ w_{pV} &= -\int_{V_i}^{V_f} p_{\mathrm{ex}}\,\mathrm{d}V, \\ p_{\mathrm{ex}} &\to p \quad \text{in the reversible limit.} \end{aligned}$$

Conditions and common mistakes / 适用条件与易错点

Writing δw is preferable to $\mathrm{d}w$: work is path-dependent and is not a state function. For a complicated moving boundary with nonuniform stresses, integrate local traction and displacement rather than using one scalar pressure.

写成 δw 比 $\mathrm{d}w$ 更能体现功依赖路径、不是状态函数。若移动边界受力不均匀或应力复杂，应对局部表面力和位移积分，而不能直接使用单一标量外压。

Quick check / 例子与检验

For expansion against a constant external pressure, $w = -p_{\mathrm{ex}}(V_f - V_i)$ even if the process is irreversible.

例如，对抗恒定外压膨胀时，即使过程不可逆，仍有 $w = -p_{\mathrm{ex}}(V_f - V_i)$。

QUESTION 25 / 27

Work as area on a pressure-volume diagram

示功图中面积与功

Is the work done during an expansion process represented by the area under the process curve on a p-V indicator diagram?

在 p-V 示功图上，膨胀功是否由过程曲线下的面积表示？

A. No / 否　B. Yes / 是

Answer and explanation / 答案与讲解

Yes, for the correct pressure curve

The integral of opposing external pressure over the volume change is geometrically the area under the p_{ex} versus V curve. This gives positive work output during expansion.

阻碍膨胀的外压对体积变化的积分，在几何上就是 p_{ex} 对 V 曲线下的面积；膨胀时这个面积表示正的输出功。

$$W_{\mathrm{out}} = \int_{V_i}^{V_f} p_{\mathrm{ex}}\,\mathrm{d}V,$$
$$w_{\mathrm{on}} = -W_{\mathrm{out}},$$
$$W_{\mathrm{out,rev}} = \int_{V_i}^{V_f} p_{\mathrm{gas}}\,\mathrm{d}V.$$

Conditions and common mistakes / 适用条件与易错点

The gas pressure curve can be used directly for reversible quasistatic expansion. In an irreversible process, gas pressure may be nonuniform or ill-defined; an equilibrium gas p-V curve between endpoints does not generally give the actual work.

可逆准静态膨胀时，可以直接使用气体压强曲线。不可逆过程中气体内部压强可能不均匀或没有明确单一数值；连接始末平衡态的气体 p-V 曲线下面积通常不是实际功。

Quick check / 例子与检验

At constant p_{ex}, the area is a rectangle of height p_{ex} and width $V_f - V_i$.

例如，外压恒定时，面积为高度 p_{ex}、宽度 $V_f - V_i$ 的矩形面积。

QUESTION 26 / 27

Diathermic versus adiabatic boundaries

透热边界与绝热边界

Is a diathermic boundary defined as one that completely prevents energy transfer as heat?

透热边界是否定义为完全阻止能量以热的形式传递的边界？

A. Yes / 是　B. No / 否

Answer and explanation / 答案与讲解

No

A diathermic boundary permits heat transfer when a temperature difference drives it. An adiabatic boundary prevents heat transfer through that boundary.

透热边界允许在温差驱动下传热；绝热边界则阻止热量穿过该边界。

$$\begin{aligned} \text{diathermic}: &\quad q \text{ may be nonzero},\\ \text{adiabatic}: &\quad q = 0 \text{ through the boundary},\\ &\Delta U = q + w. \end{aligned}$$

Conditions and common mistakes / 适用条件与易错点

A diathermic wall does not force heat to flow when the two sides are at the same temperature. An adiabatic wall does not necessarily block work or matter; those properties must be specified separately.

透热壁在两侧温度相同时不一定有净热流。绝热壁也不一定阻止功或物质通过；这些性质需要分别指定。

Quick check / 例子与检验

A thermally conducting metal wall can be diathermic, while an ideal thermos wall approximates an adiabatic boundary.

例如，导热金属壁可以是透热边界；理想保温瓶壁近似绝热边界。

QUESTION 27 / 27

Enthalpy has a plus sign

焓的定义使用加号

Is enthalpy formally defined by the relationship $H = U - pV$?

焓的正式定义是否为 $H = U - pV$？

A. No / 否　B. Yes / 是

Answer and explanation / 答案与讲解

No

Enthalpy is defined as the sum of internal energy and the pressure-volume product. It is a state function because U, p, and V are state variables.

焓定义为内能与压强—体积乘积之和。由于 U、p、V 都是状态变量，焓也是状态函数。

$$
\begin{aligned}
H &\equiv \boxed{U + pV}, \\
\Delta H &= \Delta U + \Delta(pV), \\
\mathrm{d}H &= \mathrm{d}U + p\,\mathrm{d}V + V\,\mathrm{d}p.
\end{aligned}
$$

Conditions and common mistakes / 适用条件与易错点

The expression $U - pV$ is not the definition of enthalpy. The sign choice is useful because at constant pressure and with only pressure-volume work, the $p\,\mathrm{d}V$ term cancels the expansion-work contribution in $\mathrm{d}U$.

$U - pV$ 不是焓的定义。加号之所以有用，是因为恒压且只有体积功时，$p\,\mathrm{d}V$ 恰好抵消 $\mathrm{d}U$ 中的膨胀功项。

Quick check / 例子与检验

If $U = 100\,\mathrm{J}$ and $pV = 20\,\mathrm{J}$, the enthalpy is $H = 120\,\mathrm{J}$, not $80\,\mathrm{J}$.

例如，若 $U = 100\,\mathrm{J}$ 且 $pV = 20\,\mathrm{J}$，则焓为 $H = 120\,\mathrm{J}$，而不是 $80\,\mathrm{J}$。

Formula map / 核心公式汇总

$$
\begin{aligned}
\Delta U &= q + w, \\
\delta w_{pV} &= -p_{\mathrm{ex}}\,\mathrm{d}V, \\
w_{\mathrm{rev},T} &= -nRT\ln\left(\frac{V_f}{V_i}\right), \\
H &= U + pV, \qquad \Delta H = \Delta U + \Delta(pV), \\
q_V &= \Delta U, \qquad q_p = \Delta H, \\
C_V &= \left(\frac{\partial U}{\partial T}\right)_V, \qquad C_p = \left(\frac{\partial H}{\partial T}\right)_p, \\
C_{p,\mathrm{m}} - C_{V,\mathrm{m}} &= R \quad \text{(ideal gas)}, \\
\Delta_r H &\simeq \Delta_r U + \Delta\nu_{\mathrm{g}} RT, \\
\Delta U_{\mathrm{cycle}} &= 0, \qquad q_{\mathrm{cycle}} = -w_{\mathrm{cycle}}.
\end{aligned}
$$

The q_V and q_p equations assume the work restrictions explained in Questions 6 and 7. The reaction relation assumes ideal gas behavior, the same temperature, and negligible condensed-phase pV changes.

定容热和定压热公式要求第 6、7 题所述的做功限制。反应关系要求理想气体、相同温度，并忽略凝聚相 pV 的变化。

Sources / 来源

Primary source: the displayed Notebook quiz *2 Quiz heat-work yes/no*, accessed on 29 October 2026. The seven listed source PDFs were not separately consulted.

题目主来源为当前 Notebook 中的“2 Quiz heat-work yes/no”，读取日期为 2026 年 10 月 29 日。未单独读取页面列出的七份来源 PDF。

https://notebook.google.com/notebook/982933e4-76f6-431d-b28f-bc67d4369556

[**R1**] IUPAC Gold Book, “Standard pressure” and “Standard atmosphere.”
https://goldbook.iupac.org/terms/view/S05921
https://goldbook.iupac.org/terms/view/S05906

[**R2**] IUPAC Gold Book, “Standard thermodynamic quantities” and “Standard state.”
https://goldbook.iupac.org/terms/view/S05927
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[**R3**] D. Tong, *Lectures on Statistical Physics*, section on the Debye model, University of Cambridge.
https://www.damtp.cam.ac.uk/user/tong/statphys/statmechhtml/S3.html